



Machine Learning

AIML CZG565

M1 : Introduction

Course Faculty of MTech Cluster

BITS - CSIS

BITS Pilani

Pilani Campus

Disclaimer and Acknowledgement



- The content for these slides has been obtained from books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs.
- I have provided source information wherever necessary
- I have added and modified the content flow to suit the requirements of the course and for ease of class presentation
- Students are requested to refer to the textbook and detailed content of this presentation deck over canvas

Course Introduction



- **Objective of course**
 - Introduction to the basic concepts and techniques of Machine Learning
 - Gain experience in basics of doing independent study and research in the field of Machine Learning
 - Develop skills of using recent machine learning software tools to evaluate learning algorithms and model selection for solving practical problems
- **Focus of this course**
 - Strong Mathematical Foundations of ML algorithms
 - Structured Data Analytics
 - IDD (Independent & Identically Distributed Data)
- **Topics not expected of this course**
 - Unstructured Data Analytics
 - Time Series/Sequence Data Analytics
 - Deep Learning

Course Plan

M1	Introduction
M2	Machine learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Text books and Reference book(s)

- T1 Tom M. Mitchell: Machine Learning, The McGraw-Hill Companies
- R1 Christopher M. Bishop: Pattern Recognition & Machine Learning, Springer
- P. Tan, et al. Introduction to Data Mining, Pearson
- R2 C.J.C. BURGESS: A Tutorial on Support Vector Machines for Pattern Recognition,
- R3 Kluwer Academic Publishers, Boston.

Evaluation scheme

- Quiz (10% - Best 2 of 3 quizzes)
- Assignment (20% - 1 Progressive Group Assignment)
- Mid-semester exam (30%)
- Comprehensive exam (40%)

Pre-requisites

- Linear algebra: vector/matrix manipulations, properties
- Calculus: partial derivatives
- Probability: common distributions; Bayes Rule
- Statistics: mean/median/mode; maximum likelihood

Lab Plan

Lab No.	Lab Objective
1	End to End Machine Learning
2	Linear Regression and Gradient Descent Algorithm
3	Logistic Regression Classifier
4	Decision Tree
5	Naïve Bayes Classifier
6	Random Forest

- **Labs not graded**
- **Most of the Lab recordings available at CSIS virtual labs**
- **Webinars will be conducted for lab sessions**
- **Labs will be conducted in Python**

Agenda

- What is Machine Learning?
- Why Machine Learning is important?
- Types of Machine Learning
- Application Areas
- Issues in Machine Learning
- Demo Case study

AI vs. Data Science vs. Machine Learning

Data Science

- Collection, preparation, and analysis of data
- Leverages AI/ML, research, industry expertise, and statistics to make business decisions

Data Science

Artificial Intelligence

Artificial Intelligence

- Technology for machines to understand/interpret, learn, and make 'intelligent' decisions
- Includes Machine Learning among many other fields

Machine Learning

- Algorithms that help machines improve through supervised, unsupervised, and reinforcement learning
- Subset of AI and Data Science tool

Machine Learning

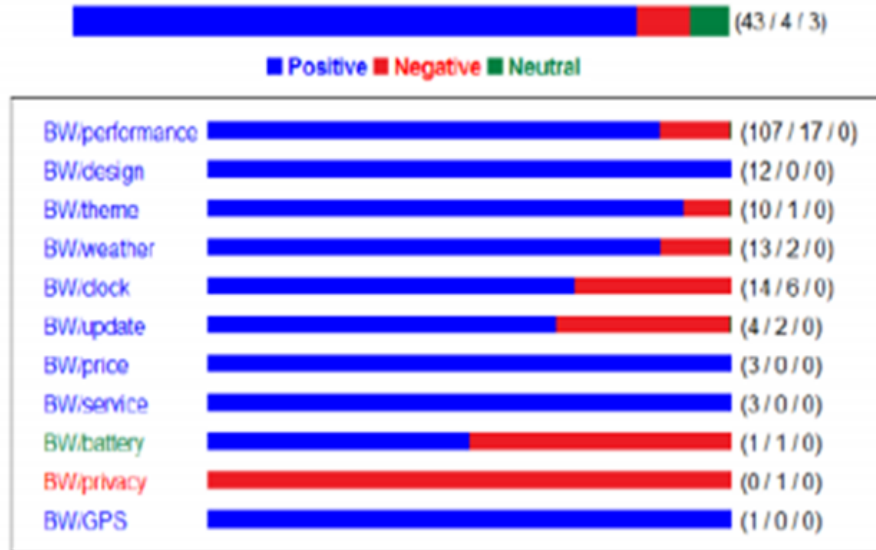
Common Use Cases

1. Applications in Smartphones
2. Transportation Optimization
3. Popular Web Services
4. Sales and Marketing
5. Security
6. Financial Domain

Introduction to Machine Learning



Common Use cases - Security & Transaction Domain



Sentiment analysis on Product review of Mobile phone



- Self Driving Cars
- Fraud Detection in Banking
- Email Filtering
- Dynamic Pricing in Travel

Derived Applications:

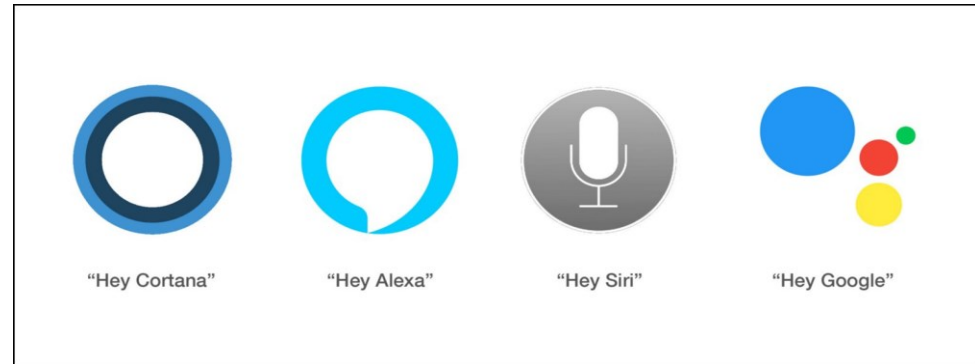
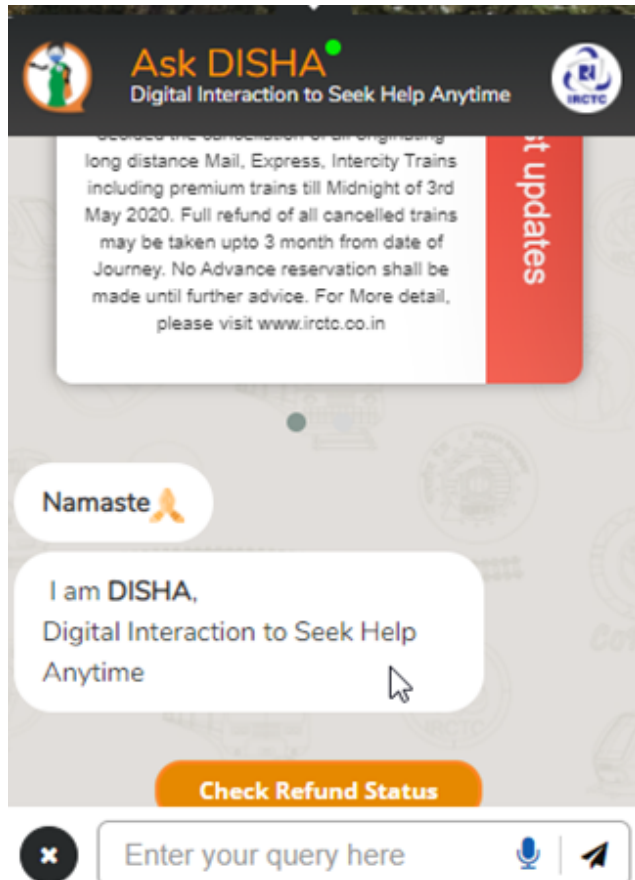
- > Cyber Security
- > Video Surveillance
- > Object Detection



Introduction to Machine Learning



Common Use cases - Customer Support Systems



- Apple's Siri
- Google Assistant
- Amazon's Alexa
- Google Duplex
- Microsoft's Cortana
- Samsung's Bixby

Derived Applications:

- > Customer Support Query (Voice vs Text)
- > Chatbots

Introduction to Machine Learning



Common Use cases - Recommendation Engines

amazon.com Recommended for You

Amazon.com has new recommendations for you based on [items](#) you purchased or told us you own.



[The Little Big Things: 163 Ways to Pursue EXCELLENCE](#)



[Fascinate: Your 7 Triggers to Persuasion and Captivation](#)



[Sherlock Holmes \[Blu-ray\]](#)



[Alice in Wonderland \[Blu-ray\]](#)

People You May Know



Maris Cohen
Communications Planner at Carat USA

Connect



Anne Michele Abbott
Graphic Designer/Art Director

Connect



Mercedes Jester
Customer Service Team Lead at John Wiley and Sons

Connect



Susan Lynch
Assistant Property Manager at The Bozzuto Group

Connect

- E-commerce sites like Amazon and Flipkart
- Book sites like Goodreads
- Movie services like IMDb and Netflix
- Hospitality sites like MakeMyTrip, Booking.com, etc.
- Retail services like StitchFix
- Food aggregators like Zomato and Uber Eats

Derived Applications:

- > Personalized Marketing
- > Personalized Banking

Additional Reading Reference for Students

Machine Learning Use Cases in real world

1. Applications in Smartphones
2. Transportation Optimization
3. Popular Web Services
4. Sales and Marketing
5. Security
6. Financial Domain
7. Other Popular Use Cases

Machine Learning Use Cases in Smartphones

Did you know that machine learning powers most of the features on your smartphone?

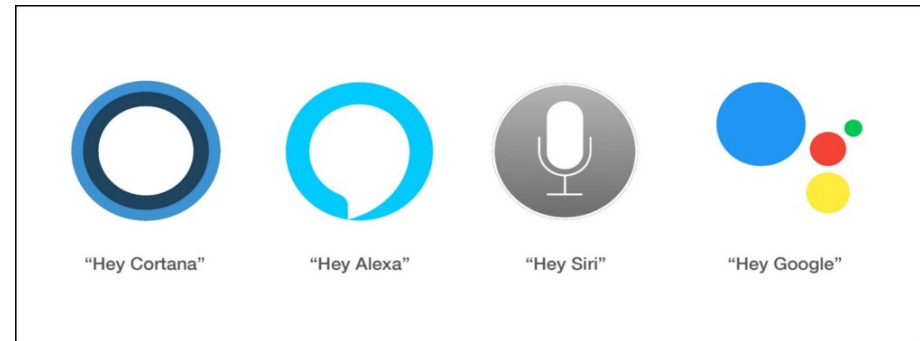
That's right! From the voice assistant that sets your alarm and find you the best restaurants to the simple use case of unlocking your phone via facial recognition – machine learning is truly embedded in our favorite devices.

Machine Learning Use Cases in Smartphones

Voice Assistants

Voice assistants are ubiquitous right now:

- Apple's Siri
- Google Assistant
- Amazon's Alexa
- Google Duplex
- Microsoft's Cortana
- Samsung's Bixby



The common thread between all of these voice assistants? They are powered by machine learning algorithms! These voice assistants recognize speech (the words we say) using Natural Language Processing (NLP), convert them into numbers using machine learning, and formulate a response accordingly.

Machine Learning Use Cases in Smartphones

Smartphone Cameras

The incredible images we are able to click these days and the depth of these images – all of that is thanks to machine learning algorithms. They analyze every pixel in a given image to detect objects, blur the background, and a whole host of tricks.

These machine learning algorithms do several things to improve and enhance the smartphone's camera:

- **Object detection** to locate and single out the object(s) (or human) in the image
- Filling in the missing parts in a picture
- Using a certain type of **neural network using GANs** to enhance the image or even extend its boundaries by imagining what the image would look like, etc.



Machine Learning Use Cases in Smartphones

Face Unlock – Smartphones

Most of us have are quite familiar with this. We pick up our smartphone and it unlocks itself by detecting our face. It's smart, efficient, time-saving and frankly superb.

What a lot of people don't know about this is that our smartphones use a technique called facial recognition to do this. And the core idea behind facial recognition is powered by – you guessed it – machine learning.

The applications of facial recognition are vast and businesses around the world are already reaping the benefits:

- Facebook uses it to identify the people in images
- Governments are using it to identify and catch criminals
- Airports are using to verify passengers and crew members, and so on



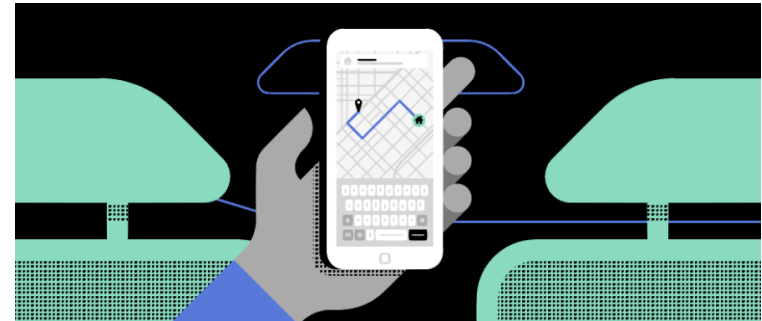
Machine Learning Use Cases in Transportation

Machine Learning Use Cases in Transportation

Dynamic Pricing in Travel

Do you often get frustrated by the surge pricing cab-hailing companies use? I encounter it on a daily basis for my commute to and from work. Prices seem to be perpetually hiked. Why is this happening?!

Imagine you're starting a ride-hailing business. You need to plan the ride prices for each route in the city in a way that would attract customers but also improve your bottomline. One way to do it is to manually map prices to each route. Not an ideal solution.



This is where dynamic pricing plays a vital role. It means adjusting your prices to changing market conditions. So prices vary depending on factors like location, time of day, weather, overall customer demand, etc. That's the underlying idea behind why surge pricing was introduced.

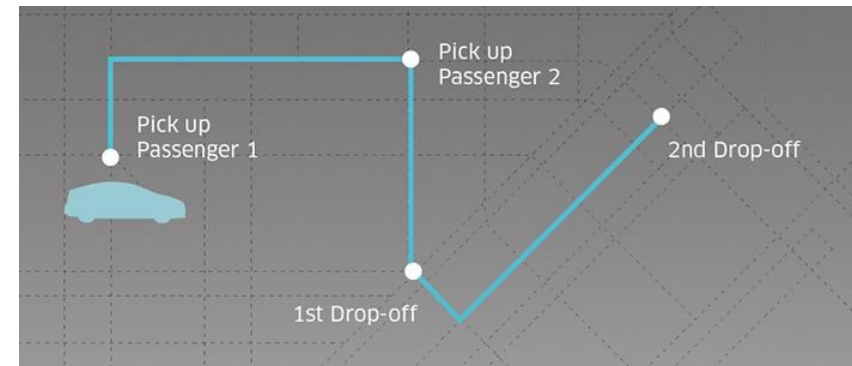
Dynamic pricing is a thriving practice in various industries, such as travel, hospitality, transportation, and logistics, among others.

Machine Learning Use Cases in Transportation

Transportation and Commuting – Uber

Dynamic pricing isn't the only machine learning use case ride-hailing companies like Uber use. They rely heavily on machine learning to identify the most optimal route to get the passenger from point A to B.

For us, it appears to be a rather simple solution. Put your location, the destination and the nearest driver will come to pick us up. But what appears to be straightforward is actually a complex web of architectures and services on Uber's backend. There are multiple machine learning techniques at play that aim to optimize the route we take.



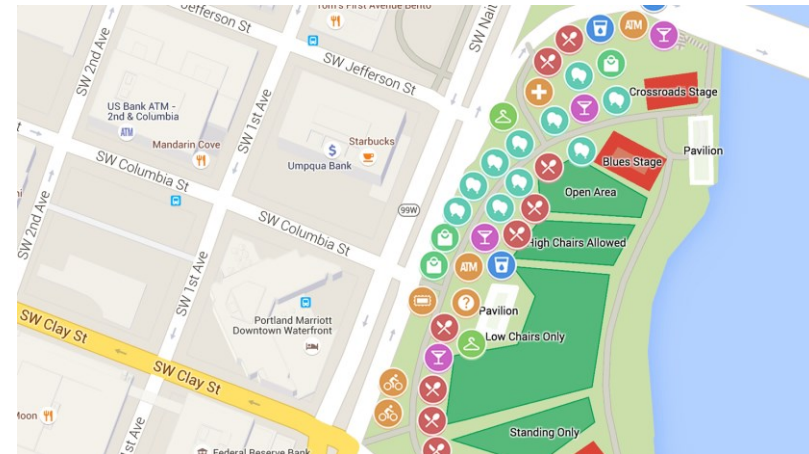
Machine Learning Use Cases in Transportation

Transportation and Commuting – Uber

Google Maps: Google Maps is a prime example of a machine learning use case

Here are some used extensively:

- Routes: Go from point A to point B
- Estimated time to travel this route
- Traffic along the route
- The 'Explore Nearby' feature: Restaurants, petrol pumps, ATMs, Hotels, Shopping Centres, etc.



Google uses a ton of machine learning algorithms to produce all these features. Machine learning is deeply embedded in Google Maps and that's why the routes are getting smarter with each update. The estimated travel time feature works almost perfectly. If it shows '40 minutes' to reach your destination, you can be sure your travel time will be approximately around that timeline. Got to love machine learning!

Machine Learning Use Cases in Popular Web Services

We interact with certain applications every day multiple times. What we perhaps did not realize until recently – most of these applications work thanks to the power and flexibility of machine learning.



Machine Learning Use Cases in Popular Web Services

Email filtering

Dealing with way too emails at work? Or is your personal email inbox bursting with utterly random and spam emails? We've all been there. My inbox count once read 11,000+ unread emails!

Wouldn't it be easy if we could write a rule that would filter emails according to their subject? A marketing mail would go to that folder. An email about work would come into my primary inbox (and so on). This would make life so much easier.

As it turns out, this is exactly what most email services are now doing! They're using machine learning to parse through the email's subject line and categorize it accordingly. Take Gmail for example. The machine learning algorithm Google uses has been trained on millions of emails so it can work seamlessly for the end-user (us).

While Gmail allows us to customize labels, the service offers default labels:

- Primary
- Social
- Promotions

The machine learning algorithms immediately categorize the email into one of these three labels as soon as you receive an email. We get an instant alert if Gmail deems it a 'Primary' email.



Machine Learning Use Cases in Popular Web Services

Google Search

The most popular machine learning use case in this (or any) list. Everyone has used Google Search and most of us use it multiple times on a daily basis. I would venture to say we take it for granted that Google will serve us the best results up front.

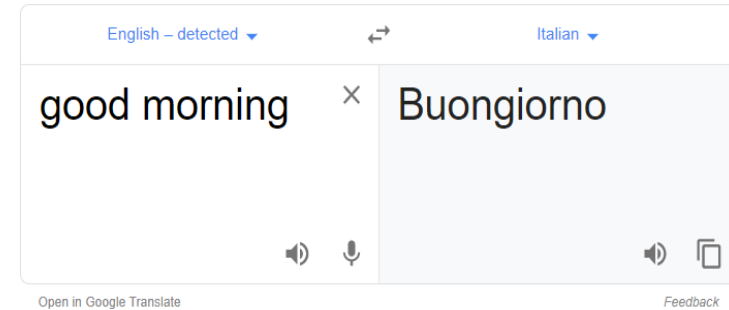
Google Search has become an impenetrable behemoth that mortals cannot crack. How it works underneath is something only those folks who have designed Google Search know. One thing we can say for certain – Google uses machine learning to power its Search engine.

The amount of data Google has to constantly train and refine its algorithms is a number we cannot fathom. No calculator in the world will tell us the number of queries Google has processed in the last two decades. It is a treasure trove for data scientists!

Machine Learning Use Cases in Popular Web Services

Google Translate

It won't surprise you to know that Google uses machine learning to understand the sentence(s) sent by the user, convert them to the requested language, and show the output.



Machine learning is deeply embedded in Google's ecosystem and we are all benefitting from that.

Machine Learning Use Cases in Popular Web Services

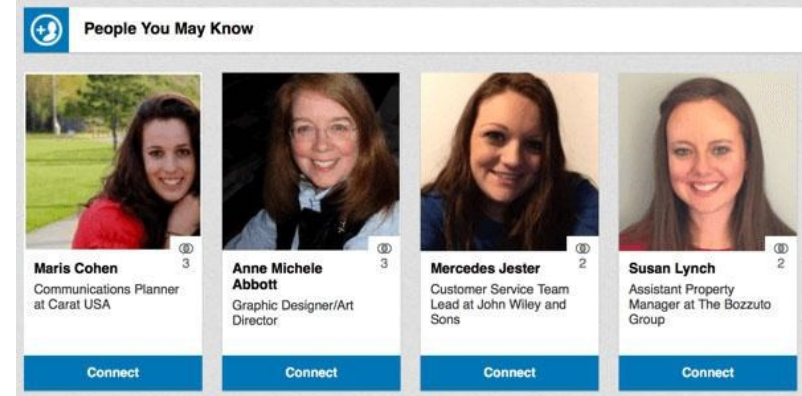
LinkedIn and Facebook recommendations and ads

Social media platforms are classic use cases of machine learning. Like Google, these platforms have integrated machine learning into their very fabric. From your home feed to the kind of ads you see, all of these features work thanks to machine learning.

A feature which we regularly see is 'People you may know'. This is a common feature across all social media platforms, Twitter, Facebook, LinkedIn, etc. These companies use machine learning algorithms to look at your profile, your interests, your current friends, their friends, and a whole host of other variables.

The algorithm then generates a list of people that match a certain pattern. These people are then recommended to you with the expectation that you might know them (or at least have profiles very similar to yours).

The ads that we see work in a similar fashion. They are tailored to your tastes, interests and especially your recent browsing or purchase history. If you are a part of a lot of data science groups, Facebook or LinkedIn's machine learning algorithm might suggest machine learning courses.



Machine Learning Use Cases in Sales and Marketing

Top companies in the world are using machine learning to transform their strategies from top to bottom.

The two most impacted functions? Marketing and Sales!

These days if you're working in the marketing and sales field, you need to know at least one Business Intelligence tool (like Tableau or Power BI). Additionally, marketers are expected to know how to leverage machine learning in their day-to-day role to increase brand awareness, improve the bottomline, etc.



Machine Learning Use Cases in Sales and Marketing

Recommendation Engines

Before the advent of IMDb (and Netflix), we all used to go to DVD stores or rely on Google to search for movies to watch. The store clerk would offer suggestions on what to watch and we took a hail mary pass by picking up movies we had no idea about.

That world is almost completely in the past now thanks to recommendation engines. We can log on to a site and it recommends products and services to me based on my taste and previous browsing history. Some popular examples of recommendation engines:

- E-commerce sites like Amazon and Flipkart
- Book sites like Goodreads
- Movie services like IMDb and Netflix
- Hospitality sites like MakeMyTrip, Booking.com, etc.
- Retail services like StitchFix
- Food aggregators like Zomato and Uber Eats

The list is long. Recommendation engines are everywhere around us and marketing and Sales departers are leaning on them more than ever before to attract (and retain) new customers.



Machine Learning Use Cases in Sales and Marketing

Personalized Marketing

- Recommendation engines are part of an overall umbrella concept called personalized marketing. The meaning of this concept is in the name itself – it is a type of marketing technique tailored to an individual's need.
- Think about this. How many calls do you get from credit card or loan companies offering their services “for free”? These calls offer the same services without understanding what you want (or don't want). It's traditional marketing that is now outdated and well behind the digital revolution.
- Now imagine if these calls or emails came highly personalized to your interests. If you're a big shopaholic and that reflects in your purchase history, perhaps the message could be about a new service offering to extend your credit line. Or if you're a machine learning enthusiast, the email could offer courses suited to your taste.
- Honestly, the potential for personalized marketing is HUGE. Machine learning helps to identify customer segments and tailor your marketing campaigns for those segments. You can regularly check how your campaign is doing through metrics like open rates, clickthrough rates, and so on.



Machine Learning Use Cases in Sales and Marketing

Customer Support Queries (and Chatbots)

- You will understand this at a very personal level if you've ever dealt with customer support (and who hasn't?). Those dreaded phone calls, the interminable wait, the unresolved query – it all adds up to a very frustrating user experience.
- Machine learning is helping remove all these obstacles. Using concepts of Natural Language Processing (NLP) and sentiment analysis, machine learning algorithms are able to understand what we're saying and the tone which we are saying it in.

We can broadly divide these queries into two categories:

- Voice-based queries
- Text-based queries

For the former, machine learning algorithms detect the message and the sentiment to redirect the query to the appropriate customer support person. They can then deal with the user accordingly.

Text-based queries, on the other hand, are now almost exclusively being handled by chatbots. Almost all businesses are now leveraging these chatbots on their sites. They remove the impediment of waiting and immediately provide answers – hence, a super useful end-user experience.

Machine Learning Use Cases in Security

Machine learning is disrupting the security industry as well! The days of traditional security, where security guards used to sit for hours on end noting down vehicle numbers and stopping suspicious folks – it's slowly being phased out.

Businesses are using machine learning to better analyze threats and respond to adversarial attacks. These use cases extend to both offline threats as well as online (bank frauds, financial threats, etc.).

Machine Learning Use Cases in Security

Video Surveillance

- Organizations globally are using video surveillance for various tasks, like detecting intruders, identifying threats of violence, catching criminals, etc.
- All of this is not being done manually, however. That would be immensely time taking. So instead, machine learning algorithms are being used for the software that is put inside these surveillance cameras.
- These machine learning algorithms use various computer vision techniques (like object detection) to identify potential threats and nab offenders.

Machine Learning Use Cases in Security

Cyber Security (Captchas)

- “I’m not a robot” – does this sentence seem familiar? We often encounter this button when a website suspects it is dealing with a machine rather than a human.
- These tests are called CAPTCHA, short for Completely Automated Public Turing test. We are asked to identify traffic lights, trees, crosswalks, and all sorts of objects to prove that we are, indeed, human.
- “Because CAPTCHA is such an elegant tool for training AI, any given test could only ever be temporary, something its inventors acknowledged at the outset. With all those researchers, scammers, and ordinary humans solving billions of puzzles just at the threshold of what AI can do, at some point the machines were going to pass us by.”
- So Google is using machine learning to make CAPTCHA even more complex to decipher. The researchers are using image recognition techniques to crack these CAPTCHAS and consequently enhance their security at the backend.

Machine Learning Use Cases in the Financial Domain

Most of the jobs in machine learning are geared towards the financial domain. And that makes sense – this is the ultimate numbers field.

A lot of banking institutions till recently used to lean on logistic regression (a simple machine learning algorithm) to crunch these numbers.

Machine Learning Use Cases in the Financial Domain

Catching Fraud in Banking

- Have you ever been a victim of credit card fraud? It's a painful experience to go through. The shock of the fraud is exacerbated by the amount of paperwork the bank asks you to fill out.
- Thankfully, machine learning is solving different layers of this process. From fraud detection to fraud prevention, machine learning algorithms are changing the way banks work to improve the customer's experience.
- The challenge with this is keeping up with the level of cyber threats. These adversaries are two steps ahead of the curve at each stage. As soon as the latest machine learning solution comes up, these attackers perfect it and build on top of it.
- machine learning has definitely helped streamline the process. These algorithms are able to identify fraudulent transactions and flag them so the bank can connect with the customers ASAP to check if they made the transaction.
- A good example is to look at the spending patterns of consumers. If a purchase does not fit in with this pattern (the amount is too high, or from a different country, etc.), then the algorithms alert the bank and put the transaction on hold.

Machine Learning Use Cases in the Financial Domain

Personalized Banking

- Another use case of recommendation engines! This one is targeted specifically for the banking domain. You must be quite familiar with personalization at this point – so think about what personalized banking could mean before you read further.
- We have read about banks targeting customer microsegments and tailoring offers to them. Personalized banking takes this concept to an entirely new level.
- The ideal personalization scenario is using machine learning to anticipate the user's need and targeting segments of each individual. As a report by BCG states:
- “Personalization in banking is not primarily about selling. It's about providing service, information, and advice, often on a daily basis or even several times a day. Such interactions, as opposed to infrequent sales communications, form the crux of the customer's banking experience.



Other Popular Machine Learning Use Cases

Self-Driving Cars

The beauty of self-driving cars is that all the three main aspects of machine learning – supervised, unsupervised and reinforcement learning – are used throughout the car's design.

few features of self-driving cars where machine learning is used:

- Detecting objects around the car
- Detecting the distance between the car in front, where the pavement is located, and the traffic signal
- Evaluating the condition of the driver
- Scene classification, among many other things.



ML – What, When, Where ?

Introduction to Machine Learning

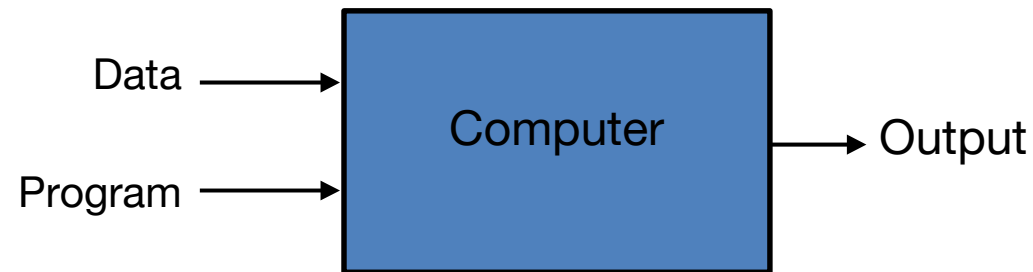


What is Machine Learning (ML)?

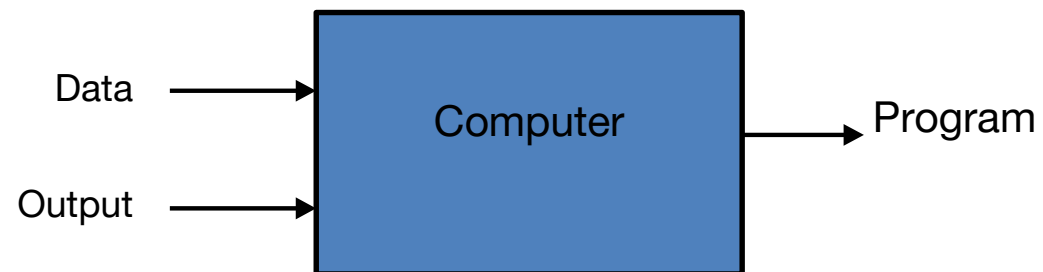
- The science (and art) of programming computers so they can *learn from data*
- More general definition
Field of study that gives computers the ability to learn without being explicitly programmed

- Engineering-oriented definition
Algorithms that improve their performance P at some task T with experience E
A well-defined learning task is given by $\langle P, T, E \rangle$

Traditional programming



Machine Learning



Slide credit: Pedro Domingos

Introduction to Machine Learning



Defining the Learning Tasks

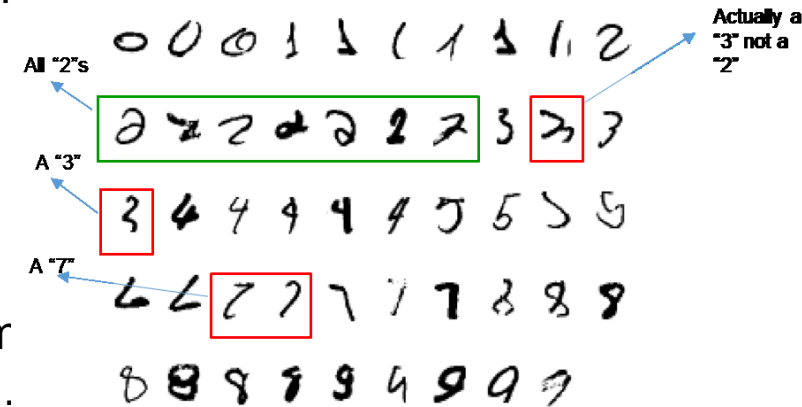
Improve on task T , with respect to performance metric P , based on experience E

Example 1

T : Recognizing hand-written words

P : Percentage of words correctly classified

E : Database of human labelled images of handwritten words



Example 2

T : Categorize email messages as spam or legitir

P : Percentage of email messages correctly classitted.

E : Database of emails, some with human-given labels

Introduction to Machine Learning



Defining the Learning Tasks

Improve on task T , with respect to performance metric P , based on experience E

Example 3

T : Playing Checkers

P : Percent of games won against opponents

E : Games Played against itself

Example 4

T : Drive on public four-lane highways using vision sensors.

P : Average distance travelled before an error (as judged by human).

E : A sequence of images and steering commands recorded while observing a human driver



Introduction to Machine Learning



Traditional Approach - Spam Filtering

Spam typically uses words or phrases such as “4U,” “credit card,” “free,” and “amazing”

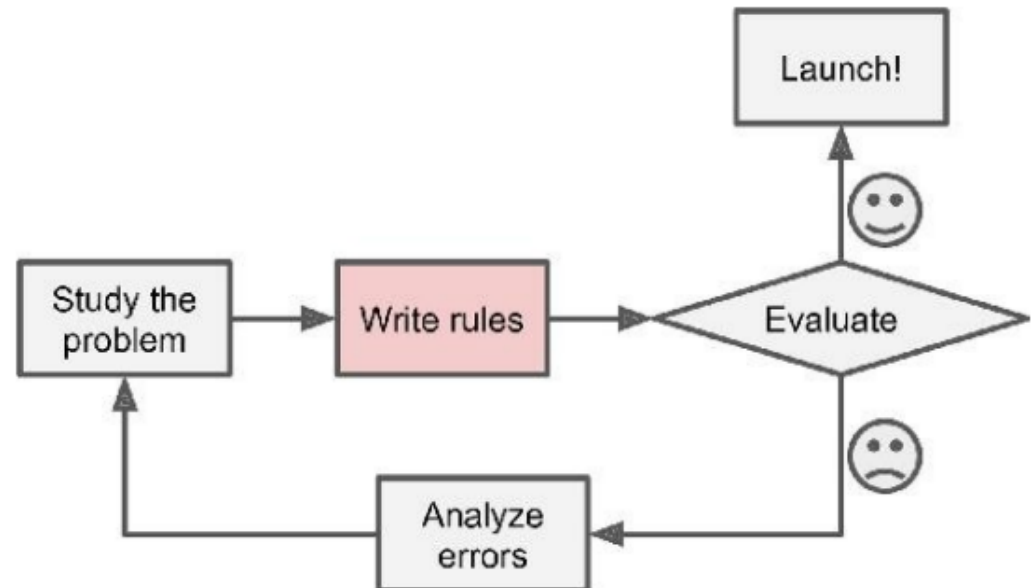
Solution

Write a detection algorithm for frequently appearing patterns in spams

Test and update the detection rules until it is good enough.

Challenge

Detection algorithm likely
to be a long list of complex rules
hard to maintain.

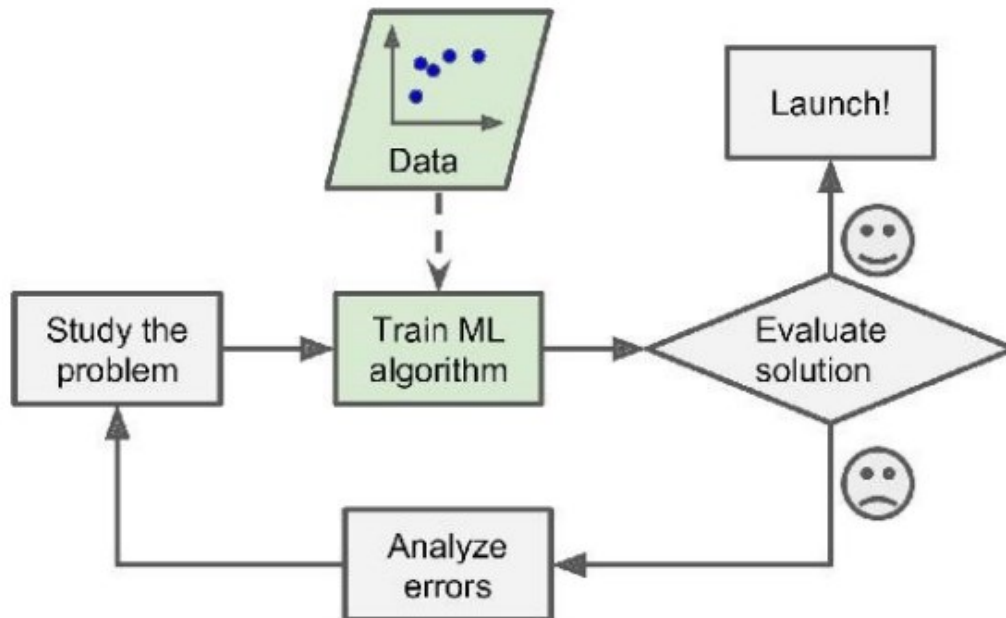


Introduction to Machine Learning



ML Approach - Spam Filtering

Automatically learns phrases that are good predictors of spam by detecting unusually frequent patterns of words in spams compared to “ham”s



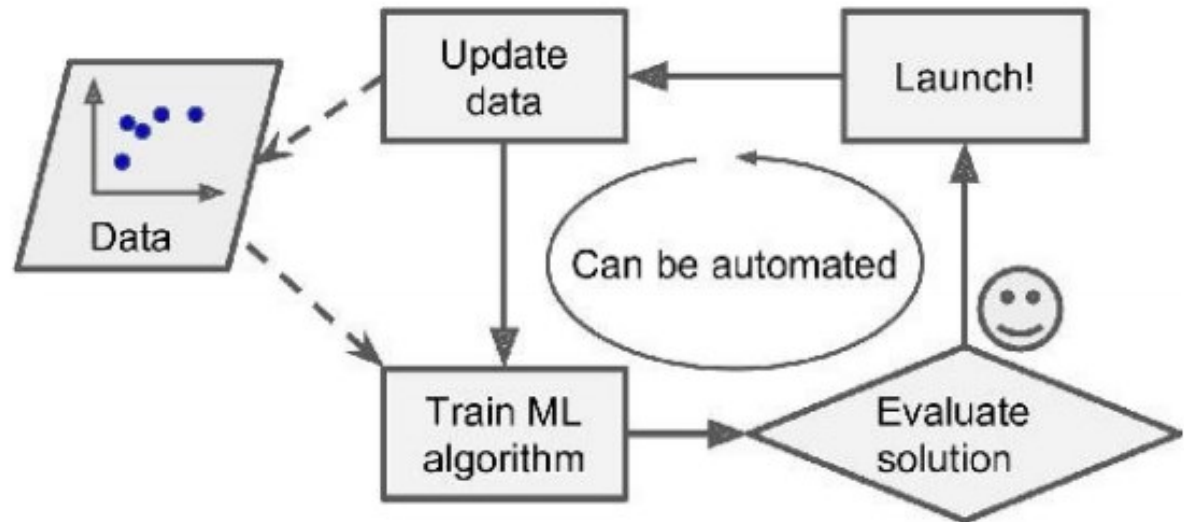
The program is much shorter, easier to maintain, and most likely more accurate.

Introduction to Machine Learning



ML Approach - Spam Filtering

Automatically learns phrases that are good predictors of spam by detecting unusually frequent patterns of words in spams compared to “ham”s



The program is much shorter, easier to maintain, and most likely more accurate.

Introduction to Machine Learning



Applications - Perspectives

Object Categorization Prediction



Medical Diagnosis
Transaction Analysis
Recommendation System
Speech – Text Processing

Sequence

Forecasting
Medical Research
Recommendation System
Content Management



Introduction to Machine Learning



Applications - Perspectives

Planning

Problem Solving

Navigation

Pattern Recognition

Path Finding

Optimization

Gaming

Decision System

Controlling

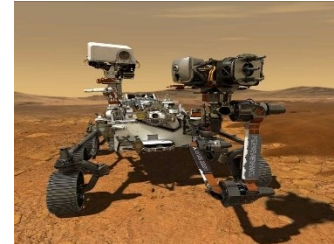


Why ML

When Do We Use Machine Learning?

ML is used when:

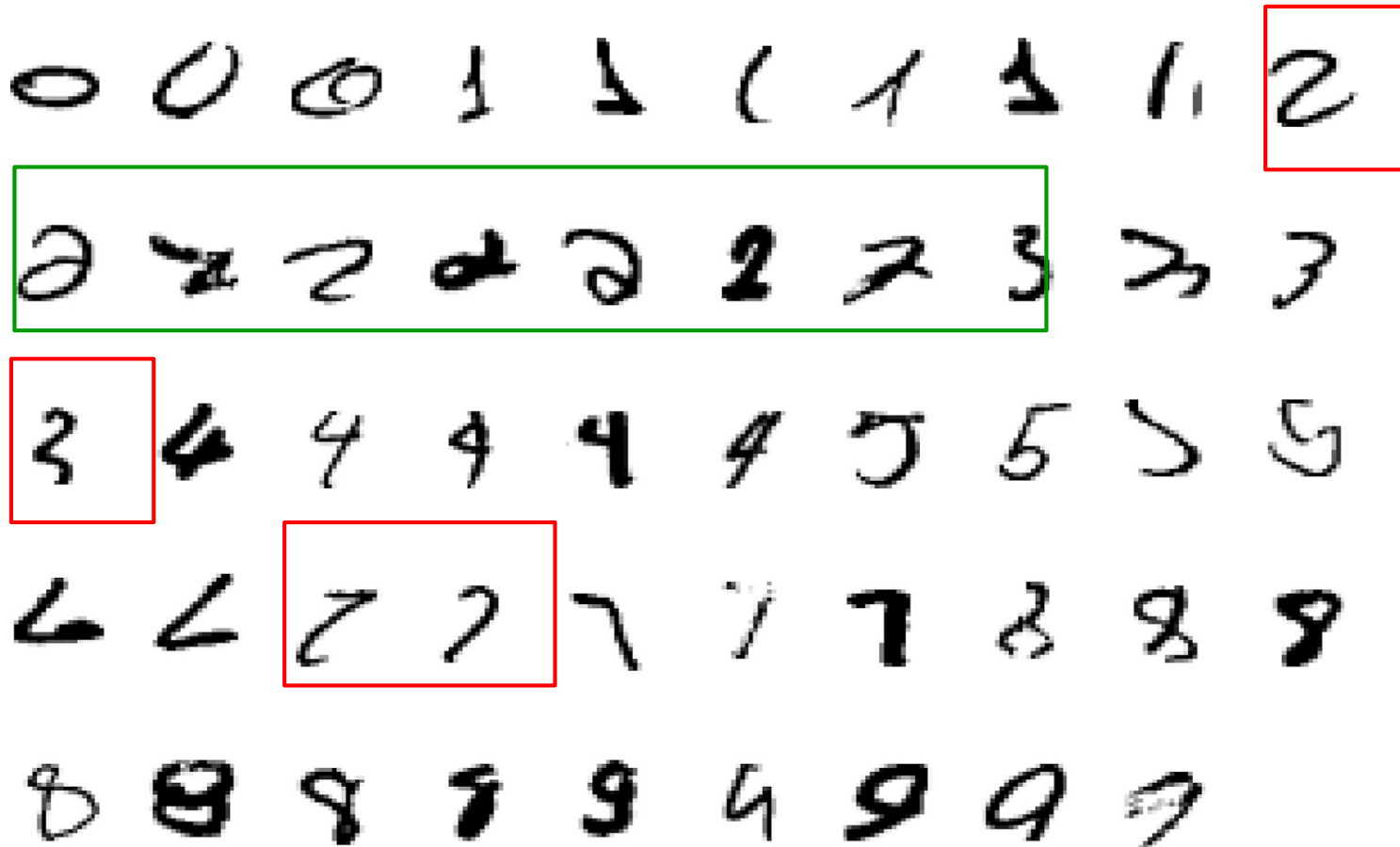
- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (Biometrics)
- Models must be customized (personalized medicine)
- **Learning isn't always useful:**
 - There is no need to "learn" to calculate payroll



Pattern recognition



It is very hard to say what makes a 2



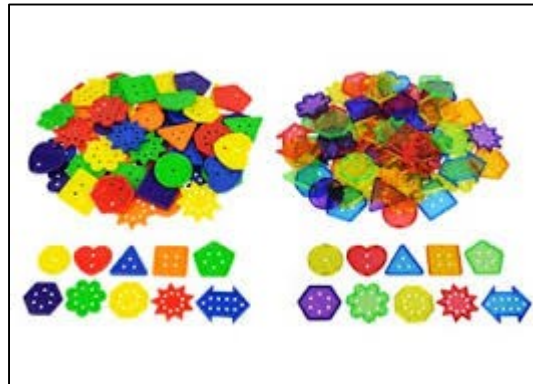
Types of ML

Introduction to Machine Learning



Types : Inputs: Based on level of supervision

Feedback	No Feedback	Delayed Feedback (rewards/penalty)
Supervised	Unsupervised	Reinforcement



Machine Learning - Examples

Objective: Employability Prediction

Features / Attributes / Predictors

- ✓ CGPA
- ✓ Communication Skills
- ✓ Aptitude
- ✓ Programming Skills

S.No.	CGPA	Communication Skills	Aptitude	Programming Skills	Job Offered?
1	9.1	Average	Good	Excellent	Yes
2	8.4	Good	Good	Good	Yes
3	8.3	Poor	Average	Average	No
4	7.1	Average	Good	Average	No
5	8.2	Good	Excellent	Excellent	No

Machine Learning - Examples

Objective : Predicting price of a used car

Features / Attributes / Predictors

- ✓ Brand
- ✓ Year (Mfg)
- ✓ Engine Capacity
- ✓ Mileage
- ✓ Distance travelled
- ✓ Cab?

S.No	Brand	Year (Mfg)	Engine Capacity	Mileage	Distance travelled	Cab?	Price (in Rs.)
1.	Honda City ZX	2008	1100	10.5	45000	N	3,50,000
2							
3							
4							

Machine Learning - Examples

Objective: Market Segmentation Study

Features / Attributes / Predictors

- ✓ Family income
- ✓ # of visits in a month
- ✓ Average money spent in a month
- ✓ Zip code

- Customers for a retailer may fall into
- ✓ two groups say big spenders and low spenders
 - ✓ three groups say big spenders, medium spenders and low spenders
 - ✓ Four groups,

S.No .	Zip Code	Family Income	# of visits in a month	Average Money Spent in a month
1	500078	11,50,000	4	8,000

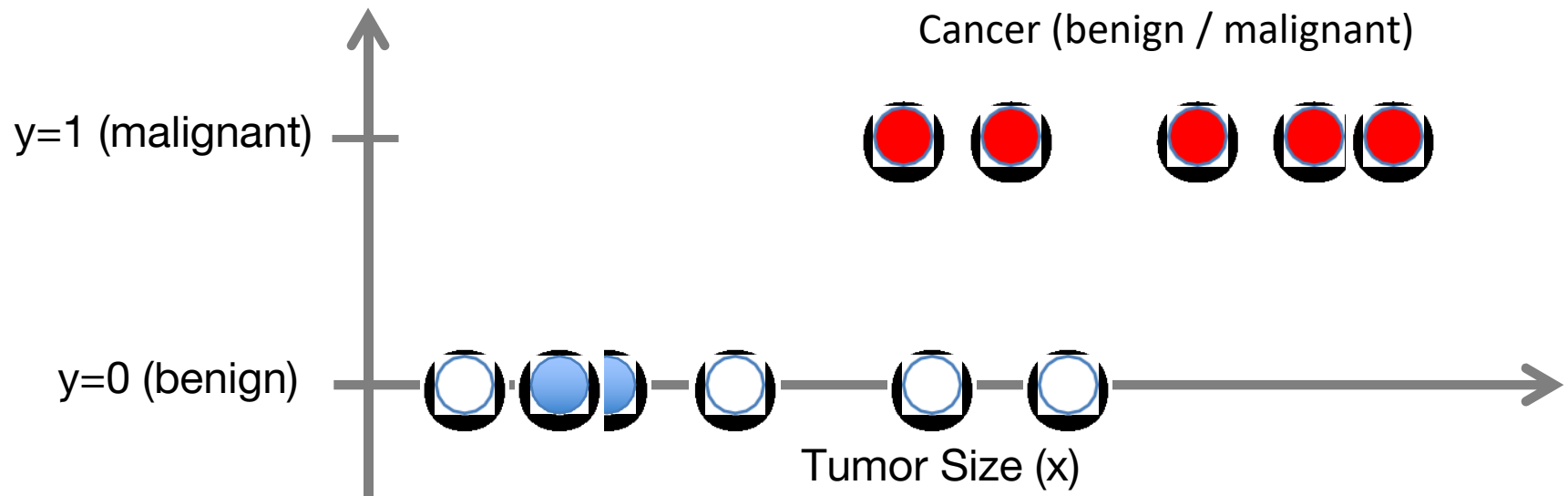
Introduction to Machine Learning



Supervised Learning: Classification

GOAL : Previously unseen records should be assigned a class as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is categorical



Slide Credit: Eric Eaton

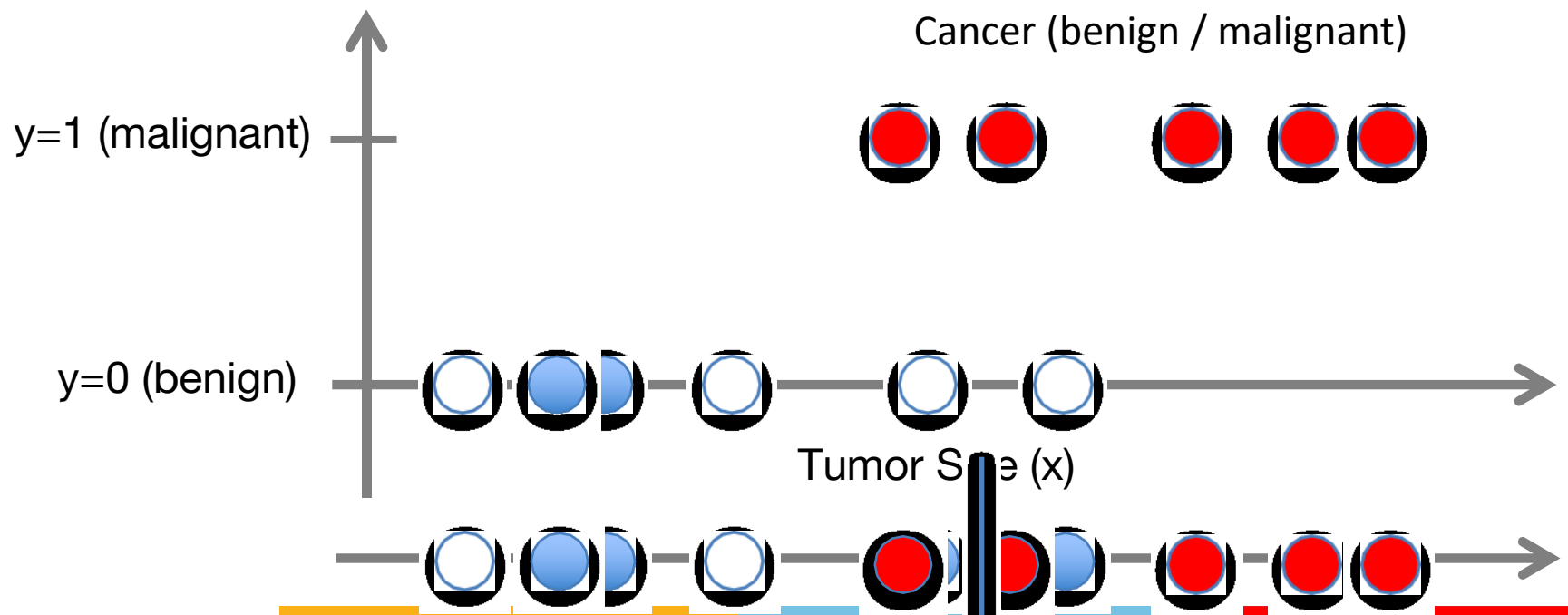
Introduction to Machine Learning



Supervised Learning: Classification

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is categorical

Learnt classifier
If $x > T$, malignant else
benign



Introduction to Machine Learning



Classification Applications

- Google Image Classification
- Face recognition system
- Spam filters – Specific Controls
- Document tagging
- Fraud detection

Introduction to Machine Learning



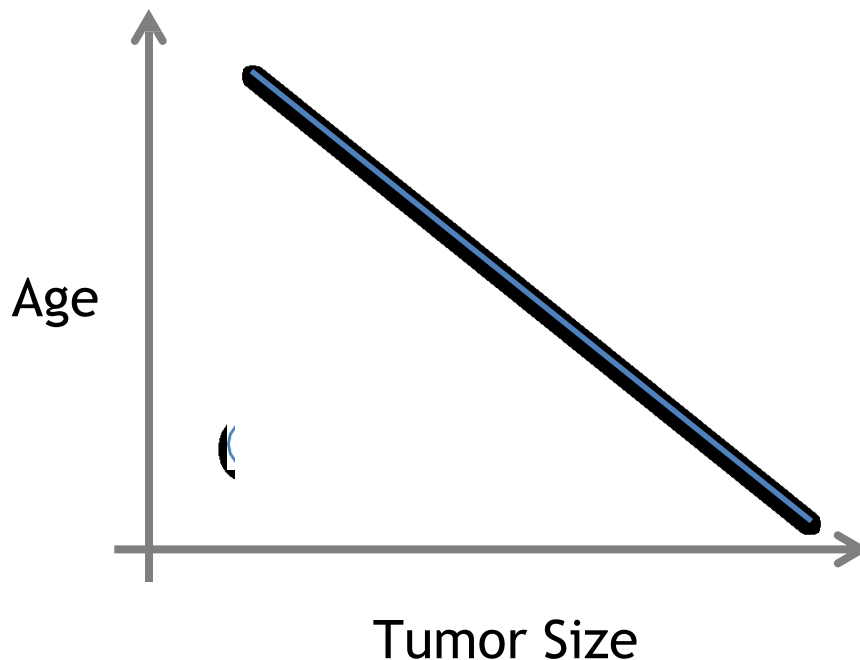
Supervised Learning: Classification

- x can be multi-dimensional
 - Each dimension corresponds to an attribute

Increasing Feature Dimension

- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

...



Introduction to Machine Learning

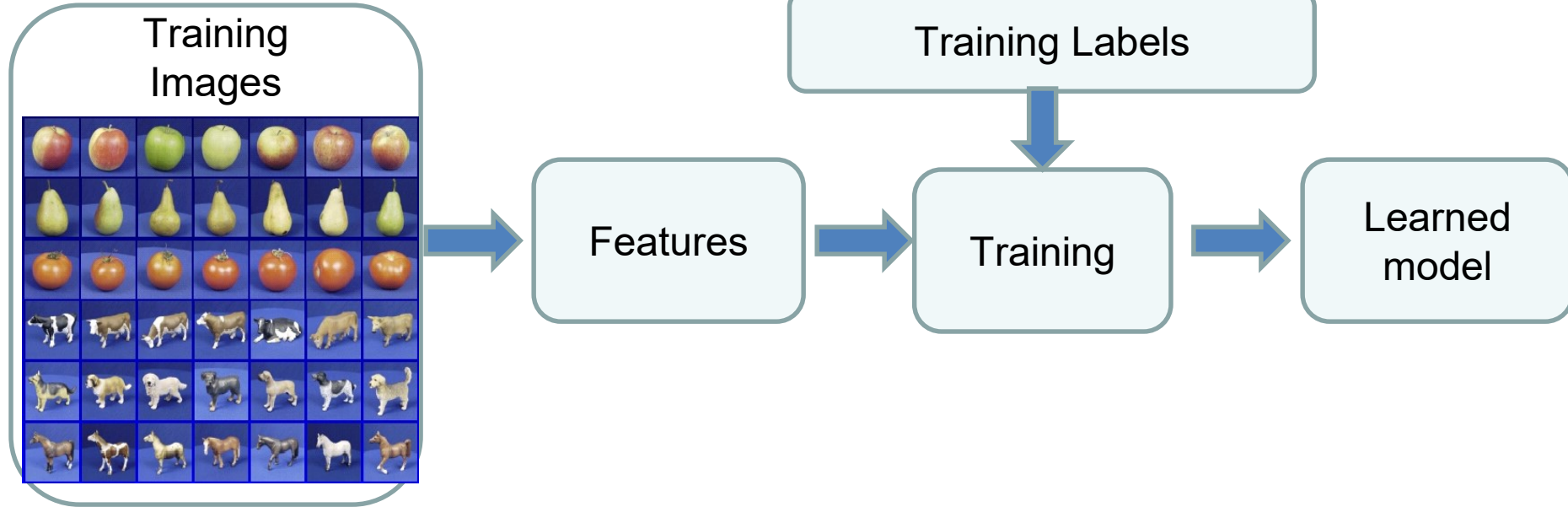


Supervised Learning Techniques / Algorithms

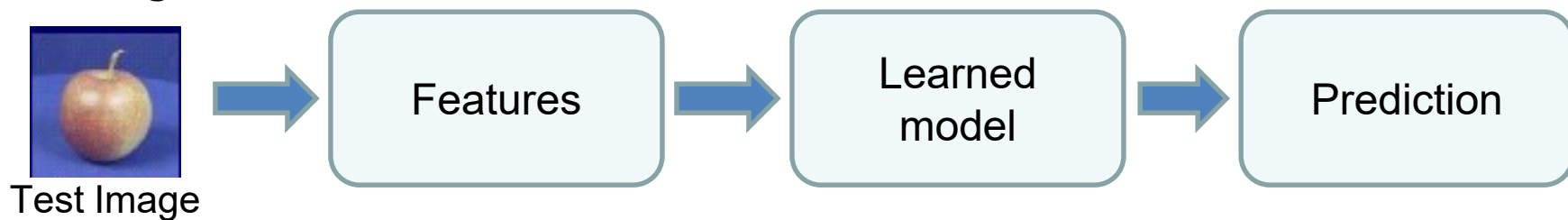
- Linear Regression
- Logistic Regression
- Naïve Bayes Classifiers
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural networks

A Typical Supervised Learning Workflow (for Classification)

Training



Testing



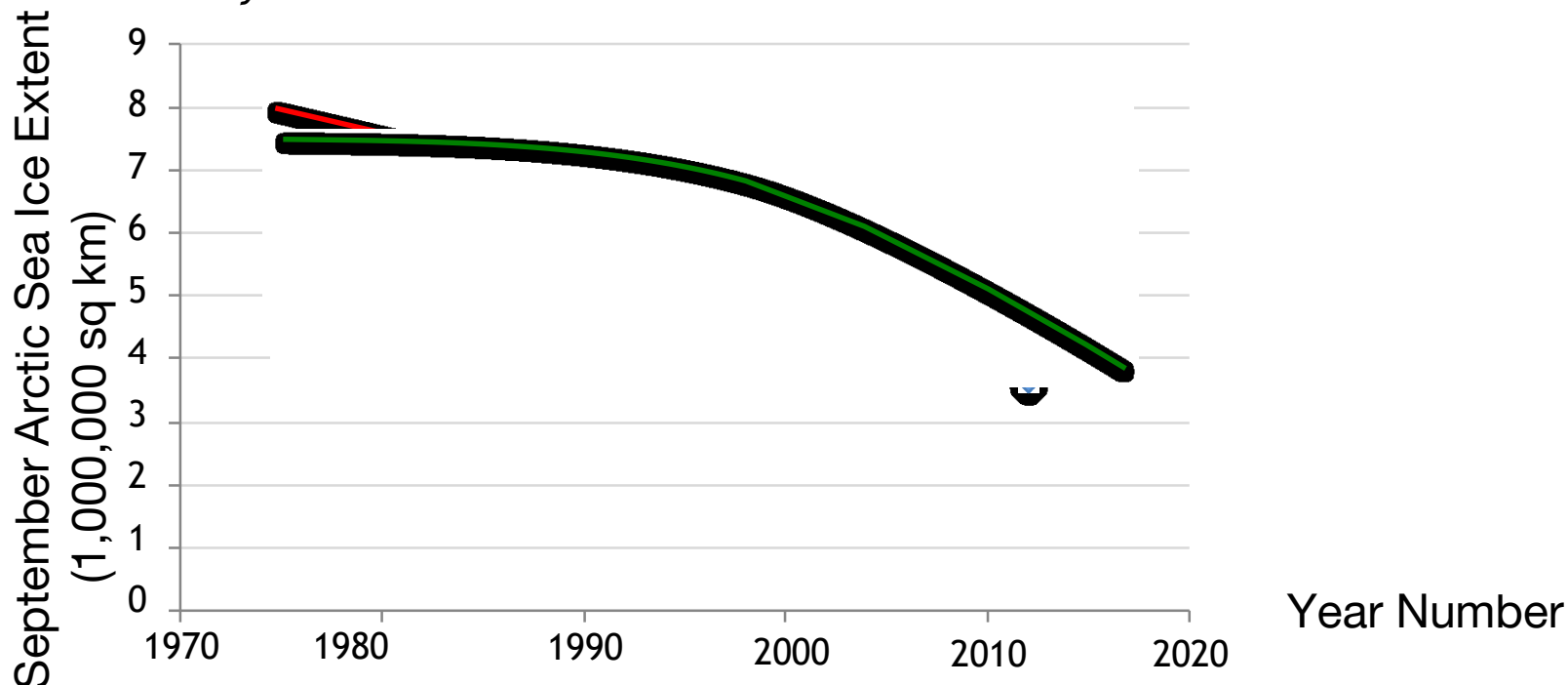
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Supervised Learning: Regression

GOAL : Previously unseen records should be assigned a value as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is real-valued



Data from G. Witt. Journal of Statistics Education, Volume 21, Number 1 (2013) Slide Credit: Eric Eaton

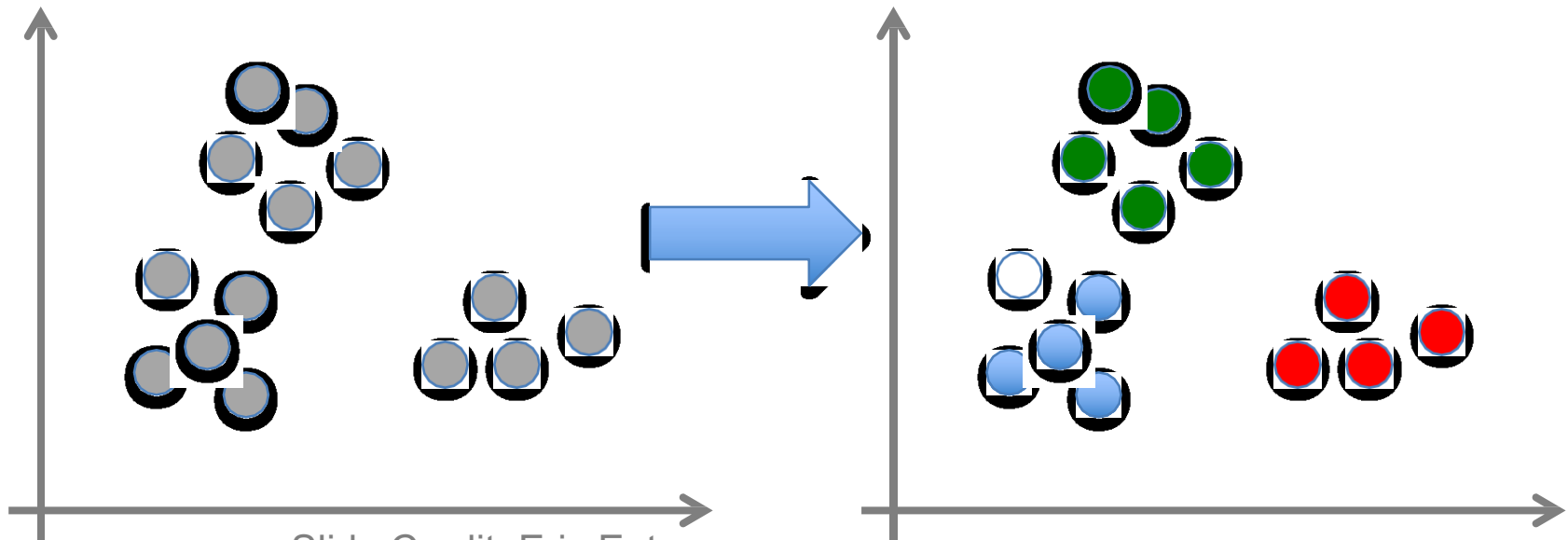
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Unsupervised Learning

GOAL : Intra cluster distances are minimized and inter cluster distances are maximized

- Given $x_1, x_2, \dots, x_n$ (without labels)
- Output hidden structure behind the x 's
 - e.g., clustering



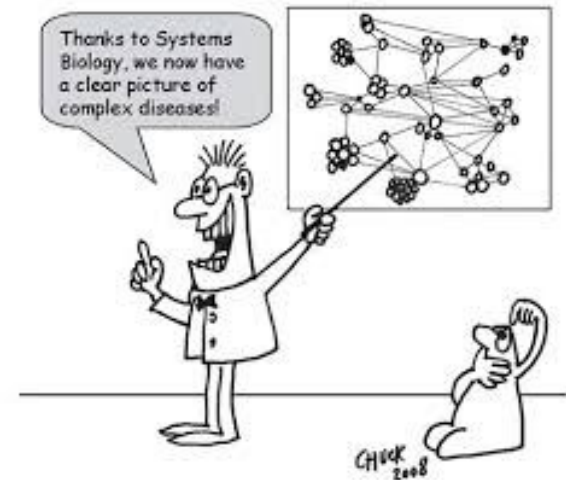
Slide Credit: Eric Eaton

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Unsupervised Learning Applications

- Personalized recommendation system
- Targeted marketing
- Spam Filters
- Content Management – News hosted in Web
- Campaigning



Unsupervised Learning Techniques

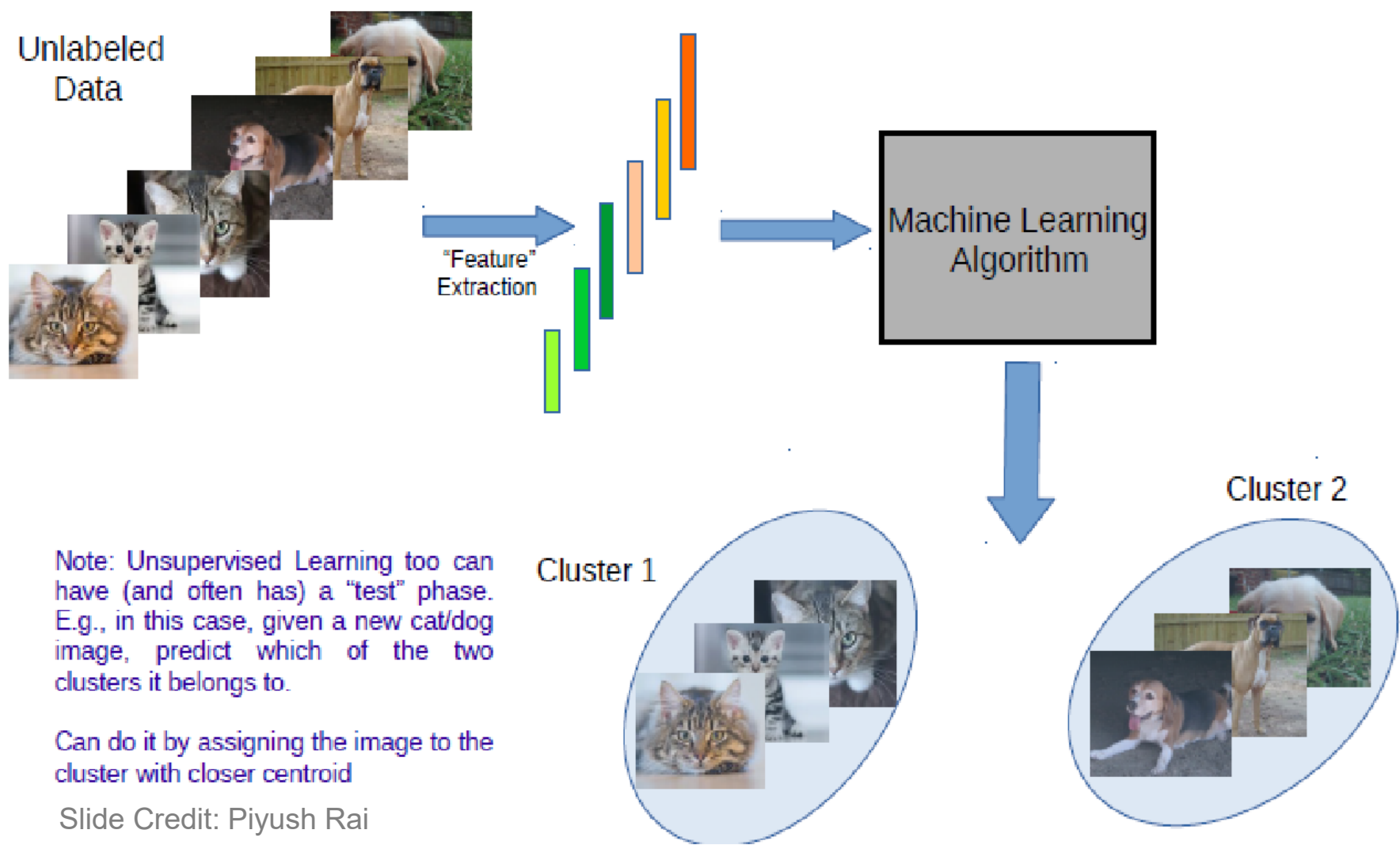
Clustering

- k-Means
- Hierarchical Cluster Analysis
- Expectation Maximization

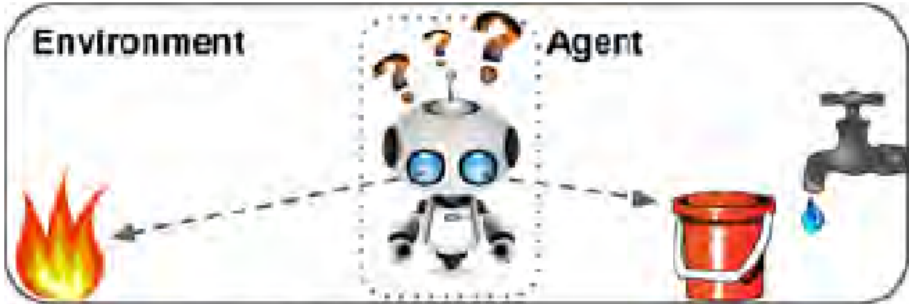
Visualization and dimensionality reduction

- Principal Component Analysis (PCA)
- Kernel PCA
- Locally-Linear Embedding (LLE)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

A Typical Unsupervised Learning Workflow (for Clustering)



Reinforcement Learning



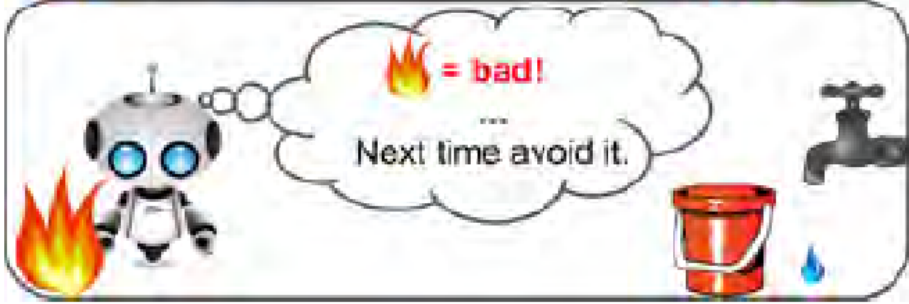
1 Observe

2 Select action using policy



3 Action!

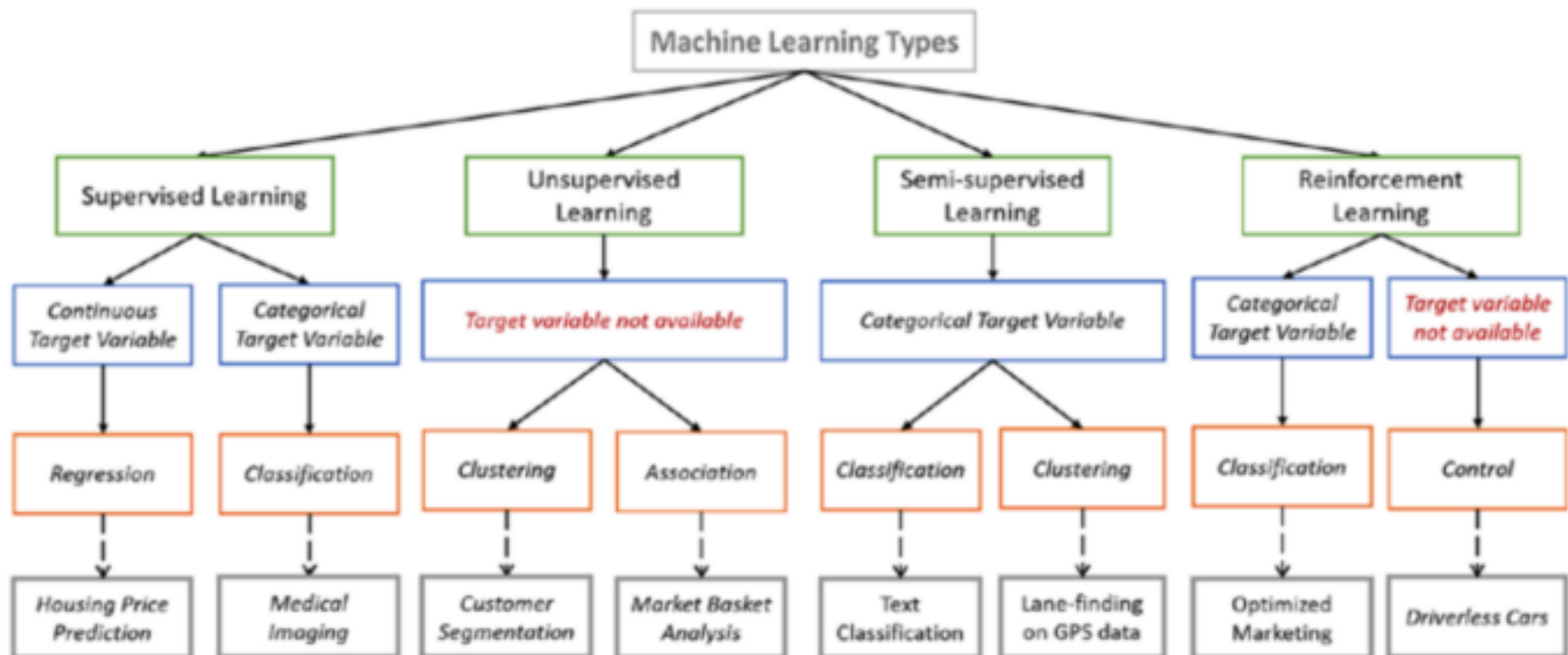
4 Get reward or penalty



5 Update policy (learning step)

6 Iterate until an optimal policy is found

Summary : Types of Learning



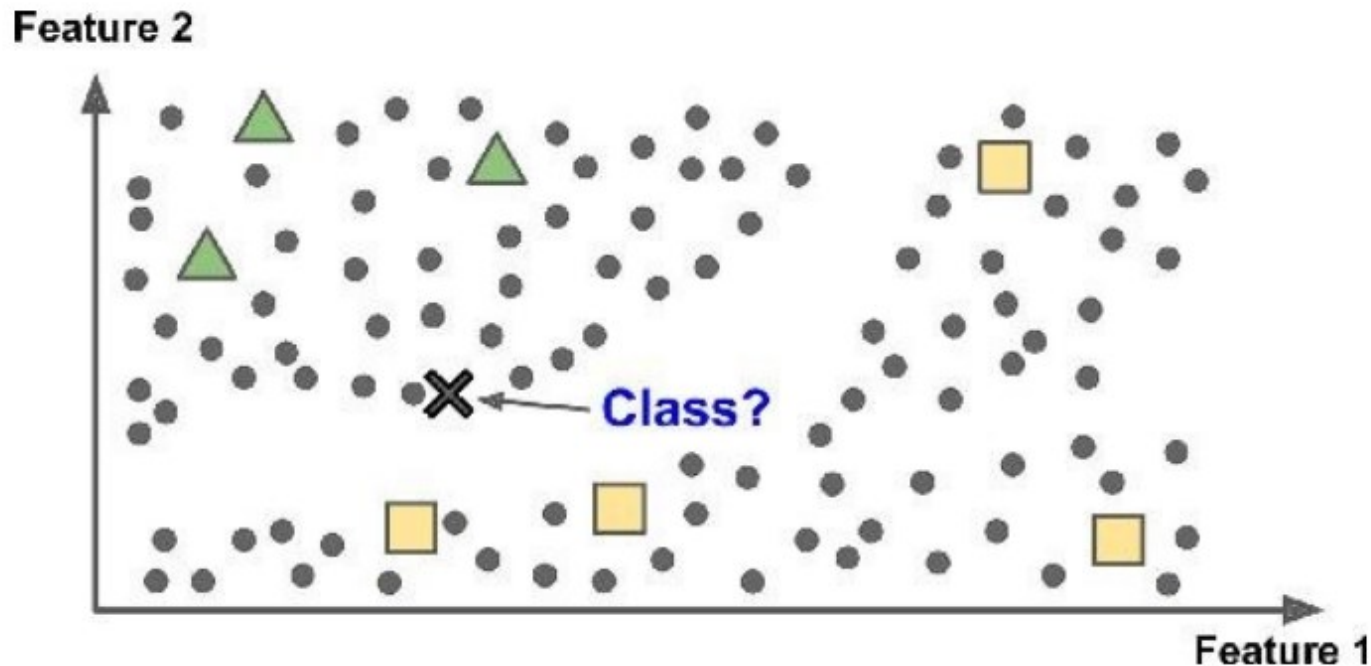
Comparison : Supervised, Unsupervised and Reinforcement Learning

Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method

Semi supervised Learning

Partially labelled data – some labelled data and a lot of unlabelled data

- Combines unsupervised and supervised learning algorithms
- Photo hosting service, e.g., google photos

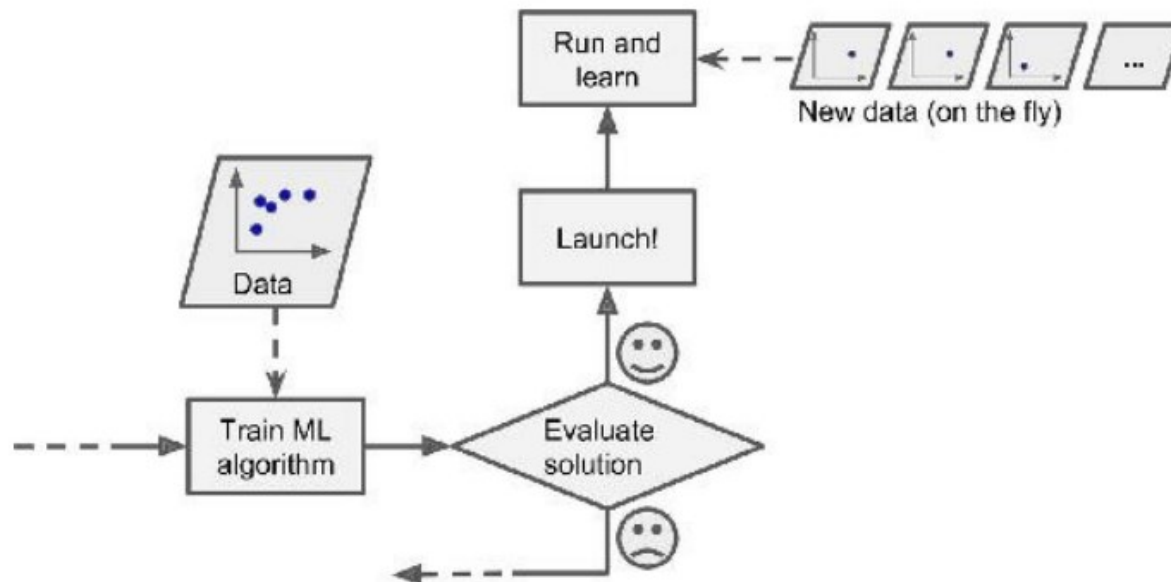


Introduction to Machine Learning



Types: Based on how training data is used

- Batch learning: Uses all available data at a time during training
- Mini Batch learning: Uses a subset of available at a time during training
- Online (incremental) learning: Uses single training data instance at a time during training

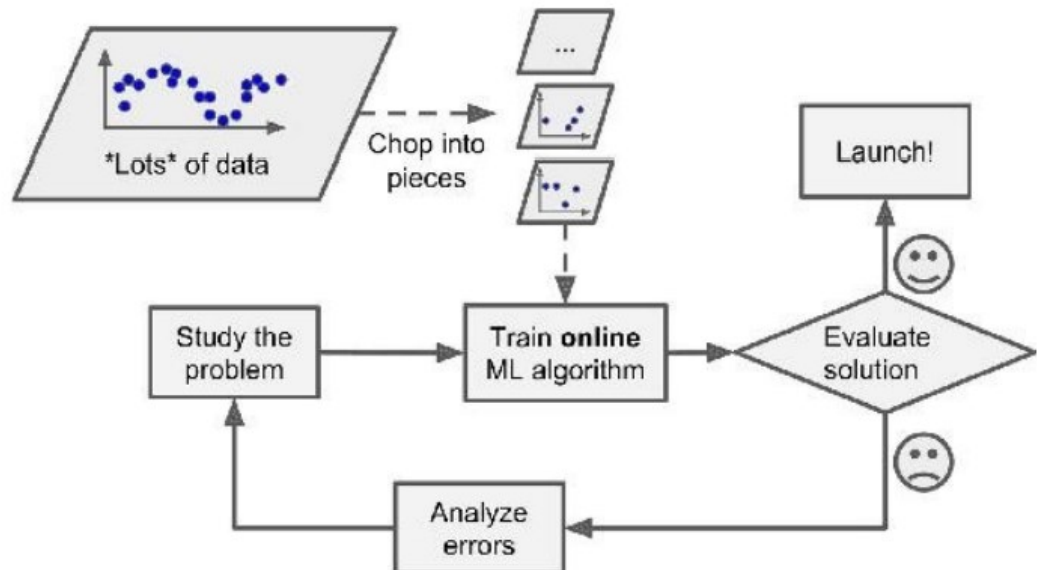


Introduction to Machine Learning



Types: Based on how training data is used

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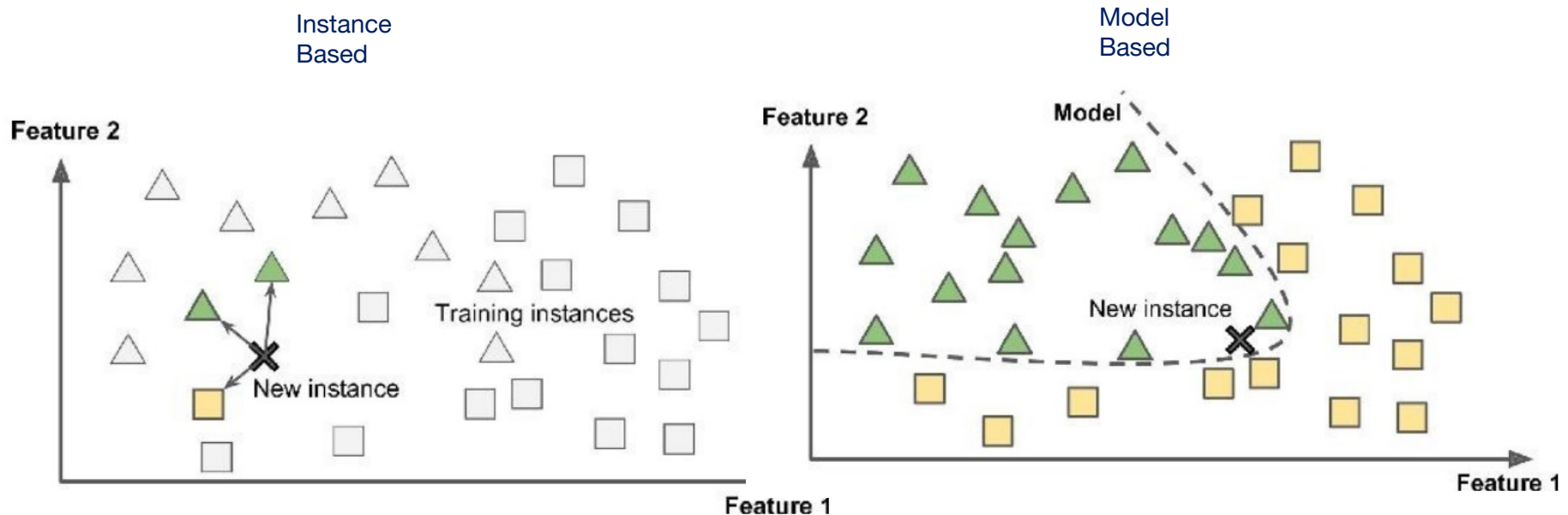


Introduction to Machine Learning



Types: Based on how training data is used

- Instance Based Learning: Compare new data points to known data points
- Model Based learning : Detect patterns in the training data and build a predictive model





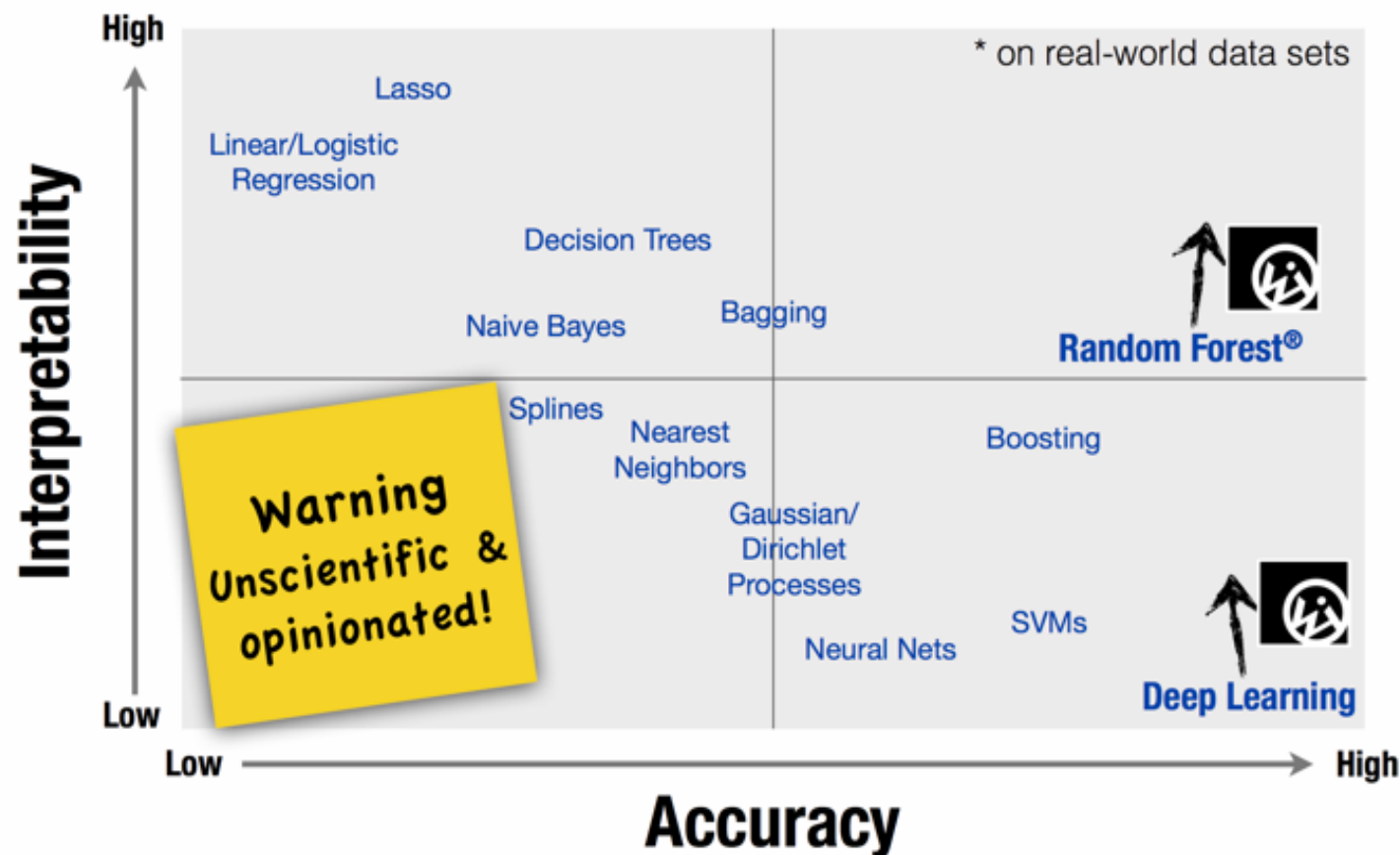
Open source ML programming tools

	Platform		Algorithms or Features
Scikit Learn	Linux, Mac OS, Windows	Python, C, C++	Classification, Regression, Clustering Preprocessing, Model Selection Dimensionality reduction.
PyTorch	Linux, Mac OS, Windows	Python, C++	Autograd Module, Optimization Module NN Module
TensorFlow	Linux, Mac OS, Windows	Python, C++	Provides a library for dataflow programming.
Weka	Linux, Mac OS, Windows	Java	Data preparation, Classification Regression, Clustering, Visualization Association rules mining

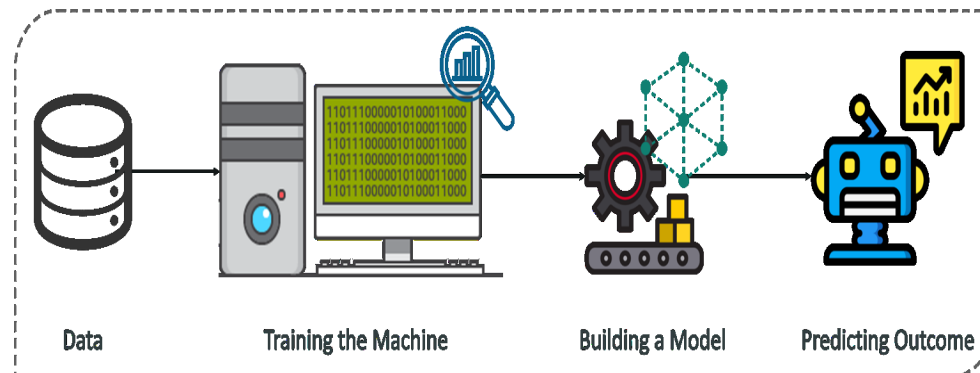
Open source ML programming tools

Colab	Cloud Service	-	Supports libraries of PyTorch, Keras, TensorFlow, and OpenCV
Apache Mahout	Cross-platform	Java Scala	Preprocessors, Regression Clustering, Recommenders Distributed Linear Algebra.
Accors.Net	Cross-platform	C#	Classification, Regression, Distribution Clustering, Hypothesis Tests & Kernel Methods, Image, Audio & Signal & Vision
Shogun	Windows Linux, UNIX Mac OS	C++	Regression, Classification, Clustering Support vector machines. Dimensionality reduction, Online learning etc.
Keras.io	Cross-platform	Python	API for neural networks

ML Algorithmic Trade-Off



ML workflow



ML workflow

1. Should I use ML on this problem?
 - Is there a pattern to detect?
 - Can I solve it analytically?
 - Do I have data?
2. Gather and organize data.
3. Preprocessing, cleaning, visualizing.
4. Choosing a model, loss, regularization, ...
5. Optimization
6. Hyper parameter search.
7. Analyze performance and mistakes, and iterate back to step 5 (or 3)

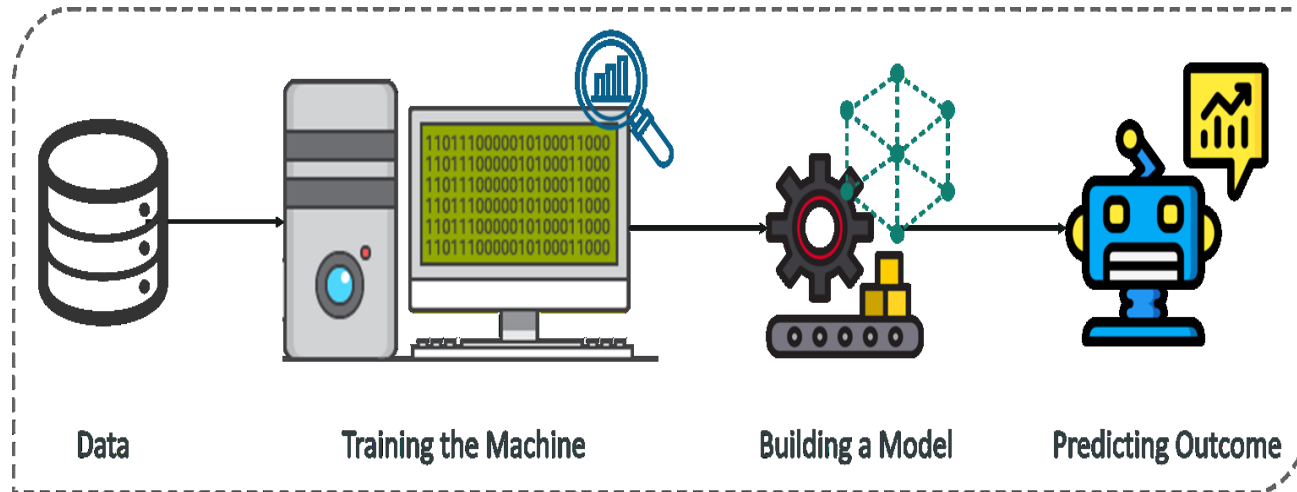
Example: Car Price prediction based on Mileage

- Define the Objective
- Data Gathering: survey, Past Purchase data
- Data Preprocessing
 - training set; test set
 - representation of input features; output
- Exploratory Data Analysis
- Choose form of model: linear regression
- System's performance evaluation: objective function
- Optimize performance by setting appropriate parameters: Optimization
- Evaluate on test set: generalization

X Mileage	Y Car Price	H0	Error L1
		9	2
		8	7
		9	3
		9	7

Additional Reading Material for Students

Machine Learning Process



A Machine Learning process begins by feeding the machine lots of data, by using this data the machine is trained to detect hidden insights and trends. These insights are then used to build a Machine Learning Model by using an algorithm in order to solve a problem.



Steps- Machine Learning Process

- The Machine Learning process involves building a Predictive model that can be used to find a solution for a Problem Statement.
- To understand the Machine Learning process let's assume that you have been given a problem that needs to be solved by using Machine Learning.
- *The problem is to predict the occurrence of rain in your local area by using Machine Learning.*

Step 1: Define the objective of the Problem Statement

- the objective is to predict the possibility of rain by studying weather conditions. Decide what kind of **data** can be used to solve this problem or the type of **approach** (algorithm) you must follow to get to the solution.

Steps- Machine Learning Process

Step 2: Data Gathering

At this stage, you must be asking questions such as,

- What kind of data is needed to solve this problem?
- Is the data available?
- How can I get the data?

Data collection can be done **manually** or by **web scraping**. There are 1000s of data resources on the web, you can just download the data set and get going.

the data needed for weather forecasting includes measures such as ***humidity level, temperature, pressure, locality, whether or not you live in a hill station***, etc. Such data must be collected and stored for analysis.

Steps- Machine Learning Process

Step 3: Data Preparation

• The data you collected is almost never in the right format. You will encounter a lot of inconsistencies in the data set such as missing values, redundant variables, duplicate values, etc. Removing such inconsistencies is very essential because they might lead to wrongful computations and predictions. Therefore, at this stage, you scan the data set for any inconsistencies and you fix them then and there. It may involve steps like **Cleaning, Formatting, Sampling, Decomposition, Scaling**

Step 4: Exploratory Data Analysis

EDA or Exploratory Data Analysis is the **brainstorming stage** of Machine Learning. Data Exploration involves **understanding the patterns** and **trends** in the data. At this stage, all the useful **insights are drawn** and **correlations between the variables** are understood.

For example, in the case of predicting rainfall, we know that there is a strong possibility of rain if the temperature has fallen low. Such correlations must be understood and mapped at this stage.

Steps- Machine Learning Process

Step 5: Building a Machine Learning Model

All the insights and patterns derived during Data Exploration are used to build the Machine Learning Model. This stage always begins by **splitting the data** set into two parts, **training data, and testing data**.

The training data will be used to build and analyze the model. The logic of the model is based on the Machine Learning Algorithm that is being implemented.

In the case of predicting rainfall, since the output will be in the form of True (if it will rain tomorrow) or False (no rain tomorrow), we can use a [Classification Algorithm](#) such as [Logistic Regression](#).

Choosing the right algorithm depends on the type of problem you're trying to solve, the data set and the level of complexity of the problem.



Steps- Machine Learning Process

Step 6: Model Evaluation & Optimization

After building a model by using the training data set, it is finally time to put the model to a test. The testing data set is used to check the **efficiency of the model** and **how accurately** it can predict the outcome. Once the accuracy is calculated, any further improvements in the model can be implemented at this stage. Methods like **parameter tuning** and **cross-validation** can be used to improve the performance of the model.

Step 7: Predictions

Once the model is evaluated and improved, it is finally used to **make predictions**. The final output can be a **Categorical variable** (eg. True or False) or it can be a **Continuous Quantity** (eg. the predicted value of a stock).

In our case, for predicting the occurrence of rainfall, the output will be a categorical variable.

References

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- Chapter 1, 2 – Introduction to Machine Learning, 2nd edition, Ethem Alpaydin
- Chapter 1 - Pattern Recognition & Machine Learning Christopher M. Bishop
- <http://www.cs.princeton.edu/courses/archive/spr08/cos511/> [Web]
- <https://www.softwaretestinghelp.com/machine-learning-tools/>

Thank you !